

$$a = r\omega^2 \left(\cos\theta + \frac{\cos 2\theta}{n} \right)$$

$$F = ma$$

$$= \underbrace{m r \omega^2 \cos\theta}_{\text{primary force}} + \underbrace{m r \omega^2 \frac{\cos 2\theta}{n}}_{\text{secondary force}}$$

- Secondary force's frequency twice more than that of primary force
- Its magnitude is $\frac{1}{n}$ times the magnitude of primary force.

$$m r (2\omega)^2 \frac{\cos 2\theta}{4n} = m r \omega^2 \frac{\cos 2\theta}{n}$$

imaginary crank	actual crank
2ω	ω
$r/4n$	r
m	m

Condition for complete balancing of reciprocating masses :

- 1) Primary force must balance
(force polygon closed)

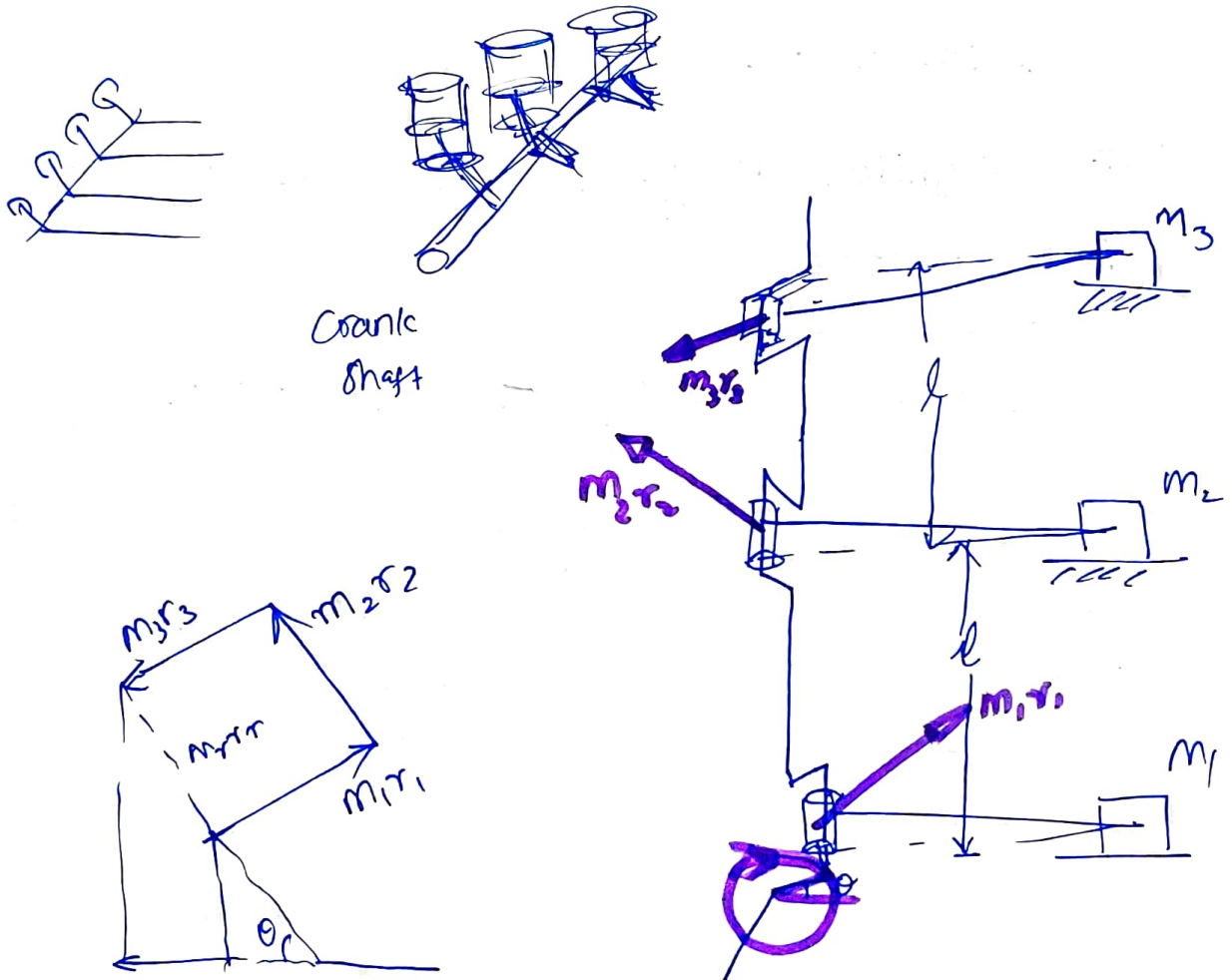
ii) Primary couple must balance

(couple polygon closed)

iii) secondary forces must balance

iv) secondary couple must balance

Balancing of inline engines



Crank
shaft

1m of stroke
secondary

$$m_r r_r \omega^2 = \text{unbalanced force}$$

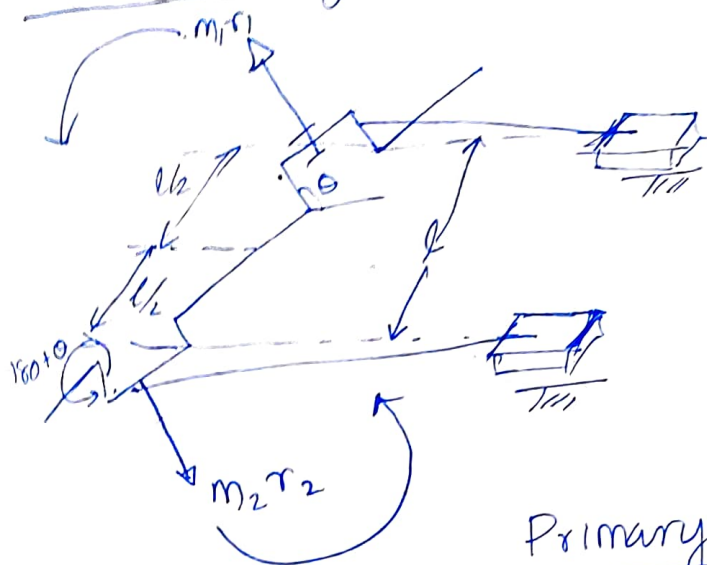
$$i \text{ force} = \sum m r \omega^2 \cos \theta$$

$$i \text{ couple} = m r \omega^2 l \cos \theta$$

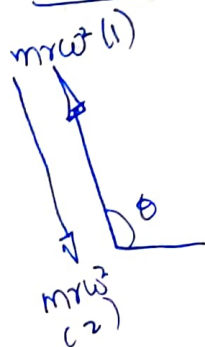
$$ii \text{ force} = \sum m r \omega^2 \frac{\cos 2\theta}{n}$$

$$ii \text{ couple} = \sum m r \left(\frac{2\omega}{4n} \right)^2 l \cos 2\theta$$

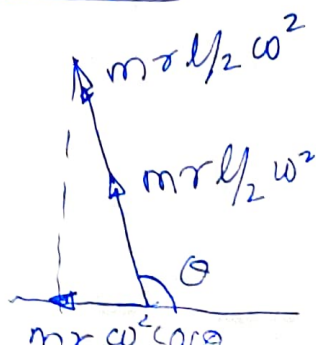
In line two cylinder engine



Primary force



Primary couple



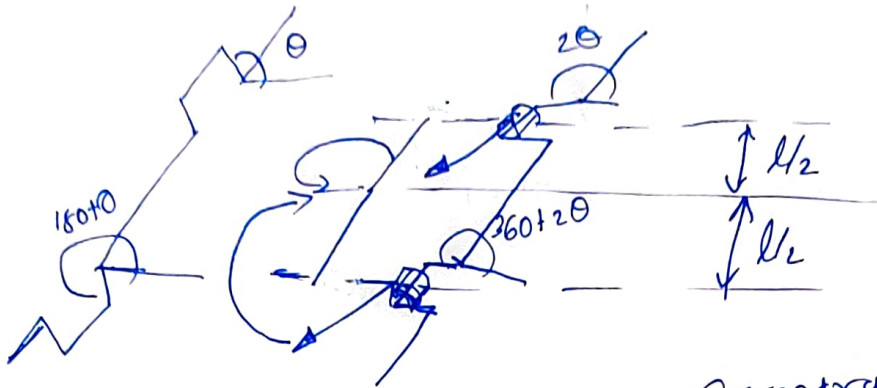
i couple

$$= m^2 \omega^2 \left[\cos \theta \frac{l}{2} + \cos(180+\theta) \frac{l}{2} \right]$$

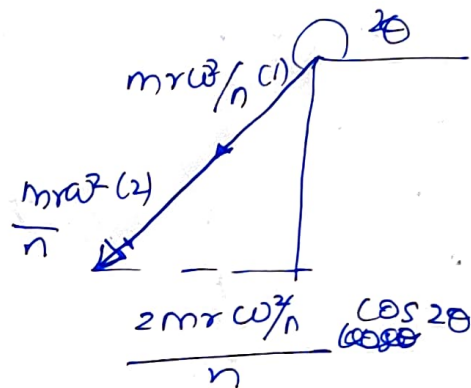
$$\text{Primary force} = m r \omega^2 (\cos \theta + \cos(180 + \theta)) = 0$$

$$\underline{\text{max}} \quad \theta = 0 \text{ \& } \theta = 180$$

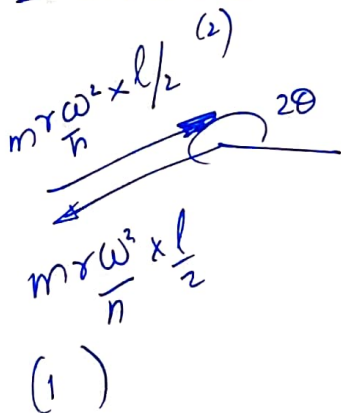
$$\theta = 90, 270$$



Secondary force polygon



couple



Secondary force

$$= \frac{m r \omega^2}{n} [\cos 2\theta + \cos(360 + 2\theta)]$$

$$\text{li force} = \frac{2 m r \omega^2 \cos 2\theta}{n}$$

secondary
couple

$$= \frac{mr\omega^2}{h} \left[\frac{l}{2} \cos 2\theta - \frac{l}{2} \cos(360+2\theta) \right]$$

$$= 0$$

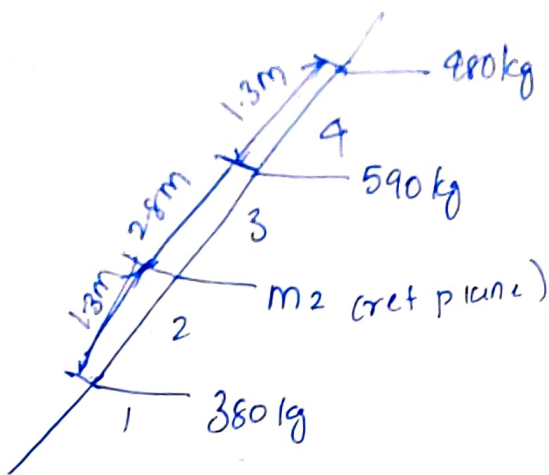
~~ii~~ ii couple = $\frac{mr\omega^2}{h} \left[\frac{l}{2} \cos 2\theta - \frac{l}{2} \cos(360+2\theta) \right]$

~~ii~~

max at $\cos 2\theta = 1$

Problem 1

A four cylinder oil engine is in complete primary balance. The arrangement of the reciprocating masses in different planes is as shown in figure. The stroke of each piston is 28 mm. Determine the reciprocating mass of cylinder 2 & the relative crank position.

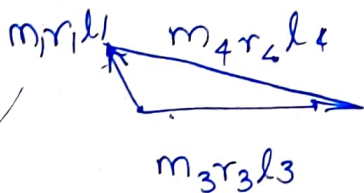
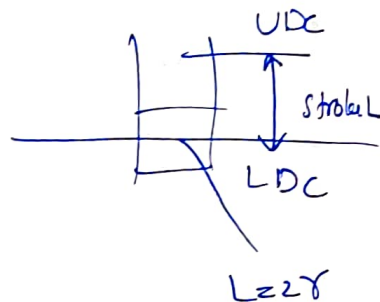
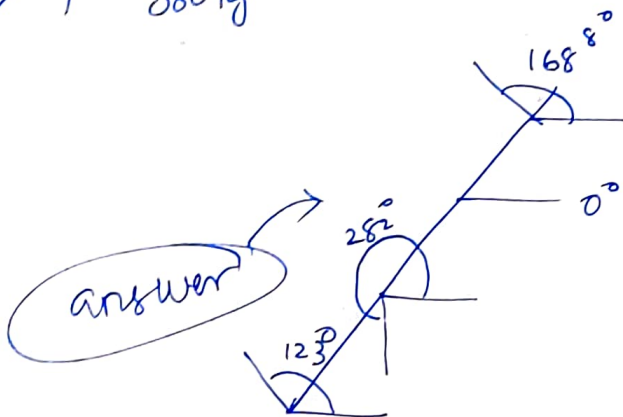


$$I = 2r$$

$$\cancel{2r} \cancel{2r}$$

$$\cancel{2r} \cancel{2r}$$

r common for all



couple polygon.

mrl

$$m_1 r_1 l_1 = 380 \times (1.3) \times r = 494r$$

$$m_3 r_3 l_3 = 590 \times (2.8) \times r = 1652r$$

$$m_4 r_4 l_4 = 480 \times 4.1 \times r = 1968r$$

force polygon

mr

mr

$$\cancel{380} 380r$$

$$590r$$

$$480r$$

$$m_1 r_1 + m_3 r_3 + m_4 r_4 =$$

$$-494r \cos \theta_1 + 1652r \cos \theta_3$$

$$+ 1968r \cos \theta_4 = 0$$

$$1652 + 1968 \cos \theta_4 = 494 \cos \theta_1 \quad \text{--- (1)}$$

$$-494 \sin \theta_1 + 1652 \sin \theta_3 + 1968 \sin \theta_4 = 0$$

$$1968 \sin \theta_4 = 494 \sin \theta_1 \quad \text{--- (2)}$$

Squaring & adding

for

$$(1652 + 1968 \cos \theta_4)^2 + 1968^2 \sin^2 \theta_4 = (494)^2$$

$$(1652)^2 + 2 \times 1652 \times 1968 \cos \theta_4 + 1968^2 \cos^2 \theta_4 + 1968^2 \sin^2 \theta_4 = 494^2$$

$$\theta_4 = 167.9^\circ \text{ or } 192.1^\circ$$

$$\theta_4 = 167.9^\circ$$

OR take ~~any~~ either value

same

$$(ii) \div (i)$$

$$\tan \theta_1 = \frac{1968 \sin 167.9}{1652 + 1968 \cos 167.9}$$

$$\boxed{\theta_1 = 123.4^\circ}$$

$$380 r \cos \theta_1 + m_2 r \cos \theta_2 + 390 \cos 0 - 1 \\ + 480 \cos \theta_4 = 0$$

$$380 \cos (123.4) + m_2 \cos \theta_2 + 590 \\ + 480 \cos (167.9) = 0$$

$$m_2 \cos \theta_2 = -\frac{88.5}{\cos \theta_2} \quad \text{--- iii}$$

$$\uparrow \quad 380 \sin (123.4) + m_2 \sin \theta_2 + 590 \sin 0 \rightarrow 0 \\ + 480 \sin (167.9) = 0$$

$$m_2 \sin \theta_2 = -417.9 \quad \text{--- iv}$$

$$iii^2 + iv^2$$

$$m_2^2 = (88.5)^2 + (-417.9)^2$$

$$\boxed{m_2 = 427.1 \text{ kg}}$$

iv/
iii

$$\tan \theta_2 = \frac{-417.9}{88.5}$$

$$\theta_2 = 282^\circ$$